New rules, old habits: across seven team sports, each league keeps its own way of using the rules

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Abstract

Rule changes in team sports are usually judged by their average effect. By that measure, each new rule did what it was written to do. But each league used its rules in its own way, and kept doing so. We measured this in 256,966 matches, games and innings in seven sports. The changes included a higher limit on substitutions, an extra time-out, a new overtime format, a longer three-point line and a pitch clock. When the substitution limit rose, the leagues were already in an order, from most substitutions to fewest. Each league added its own number of substitutions. But they kept that order from their first season under the new limit to 2025–26 (rank correlation 0.73, where one means the same order and zero means unrelated orders). In most tests, leagues also kept their order under the other rules. In every sport measured, a league stayed near its place among its sport's leagues from season to season. Coaches who moved carried part of their old team's distance from its league's average (rank correlation 0.30). Of eighteen predictions written before their data were read, thirteen held. Rule-makers should judge rule changes league by league, over several seasons.

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Word count: 3,240 words of main text, from the Introduction to the Conclusion with headings, excluding tables, references and figure captions.

1 Introduction

Every team sport has rules on who may play and when. Football caps its substitutions, volleyball the players a team may name on its match sheet, handball its team time-outs and cricket the overs one bowler may bowl. Basketball and ice hockey let players come and go, and baseball now times its pitchers. These rules change often [1, 2]. Yet a change is usually judged by one number: its average effect on results, on running, on injuries or on the length of a game [3, 4, 5, 6]. An average over leagues hides how each league responds. This paper asks what each league does with the same rule, in seven sports.

A rule can say what teams may do, and we call this the room it gives them. They choose what to do with that room. A speed limit is the everyday example. It sets how fast drivers may go. Each

road then settles at its own speed below the limit. In the same way, each league uses a rule's room in its own way. We call a league's own way of using the rules its habit. A habit is measured from records the leagues already keep, such as substitutions per match or the number of players a team uses. The word describes only what teams do. A league's habit is the sum of its coaches' decisions. If habits differ, and keep differing, a rule-maker who reads the first season's average misjudges the leagues at both ends.

Football offers the cleanest case. In the summer of 2020 dozens of football leagues raised the substitution limit from three to five. The change was made for the pandemic's crowded calendar. So one new permission reached many leagues at once. IFAB, football's rule-maker, later made it permanent in the Laws of the Game [7]. Every substitution is recorded with its minute, so what each league did with the new room can be counted.

Studies of the change have followed one league at a time, or compared a few leagues and combined their response [5, 8, 6]. Leagues are also known to keep their own profiles, for example in home advantage and in fouls and cards [9, 10]. Analysts have shown that the largest football leagues kept the same order in their use of substitutes from one season to the next [11, 12]. One of them called these differences lasting and explained them by each league's customs [11]. This paper tests whether leagues keep their order through a shared rule change, in football and in six other sports. We wrote most of these tests down before reading their data. We also wrote down eighteen predictions for seasons, leagues and competitions whose records we had not yet read.

We tested four claims. First, under five substitutions football's leagues would use the new room unevenly and keep their order. Second, leagues in other sports would also keep their order in how they use their rules. Third, a league's place among its sport's leagues would persist from season to season about as strongly in every sport. Fourth, a coach who moved would carry part of the old team's distance from its league's average. If these held, a rule change would be best judged league by league, and over several seasons.

2 Methods

2.1 Sports, leagues and records

Table 1 lists the sports, the rule change studied in each, and their leagues and records. Football's records come from transfermarkt-datasets, a public collection built from Transfermarkt, a football statistics website. We read its snapshot of 3 August 2026 [13]. For each match the records carry the starting line-ups, the benches and the minute of each substitution. The fourteen European leagues whose substitutions are counted hold 43,159 matches from 2015–16 to 2025–26. Other leagues enter some measures (Table 1).

Each league's limit in each season, three or five, was taken from the competition's regulations (Supplementary Table S1). A season in which the limit changed is split at the date of the change. We count substitutions per team-match, which is one team's side of one match. Supplementary Section S1 shows how the counts were checked against an independent record, and corrected where

the records fell short.

The records of basketball, volleyball, handball, ice hockey and baseball come from the leagues' and federations' own sites. Cricket's T20 leagues play matches of twenty overs a side. Their records come from Cricsheet, a public archive of ball-by-ball records [14].

2.2 Ethics

The study reused only public match records that already existed. No one was recruited, contacted or treated. No medical records were seen, so informed consent did not apply. The records name professional players and managers, so they were treated as personal data despite being public. Processing was for scientific research under the research provisions of the General Data Protection Regulation. Every published table and figure is aggregated to leagues, or to summaries across managers, so none describes a single player or manager. Ethical approval was not sought. The author treated these published records as archival data, which Trinity College Dublin's research ethics guidance exempts from approval [15]. The same guidance grants no approval afterwards.

2.3 What is measured

A rank correlation, or Spearman's, shows how well two orders agree: one for the same order, zero for no link and minus one for the reverse.

The spread between leagues in a season is the standard deviation of their means: roughly, a typical league's gap from their mean. A league's place on a measure is how many spreads its value sits above or below the mean of its sport's leagues that season. The persistence ϕ is the share of its place a league keeps from one season to the next. So after k seasons, a league keeps on average ϕ^k of its place. We measured one ϕ for each sport. We used the correlations between every pair of seasons, with the chance swings of single games taken out (Supplementary Section S6).

A team's distance from its league is its substitutions per team-match minus its league's mean that season. We average it over each coach's spell. Basketball's distance uses the five busiest players' share of minutes, and baseball's the players used beyond the starters. The coach's carry is the rank correlation between the old team's distance and the new team's, across the coaches who moved.

2.4 Pre-specified tests and statistics

Each test was written down as a short text, which we call a pre-specified test. It fixes what the test measures, its direction, which leagues qualify and what each outcome would be called. Each was saved, with its date, in the project's online repository (Supplementary Section S4). Of football's 26 pre-specified tests, thirteen were saved before their outcome was recorded. The other thirteen were written down only when they were run.

The other sports were tested on a split record. Each game went to one of two halves by a fixed rule that works like a coin toss. Half A was read freely, to choose each measure and pass mark.

The tests were then saved before half B was read. The split guards against choosing a measure on the luck of single games. But both halves share the same leagues, so a measure chosen to suit those leagues could pass in both. The strongest evidence is therefore the eighteen predictions for new seasons, leagues or competitions. Each was written down before any of its records was read, and each was read once.

A 95% interval gives the values that the data leave plausible (Supplementary Section S1). P is the chance that luck alone gives a result at least this strong. Most tests are one-sided, because each fixed its direction in advance. A link in the other direction counts as no support. Every analysis added in revision is labelled stated: it was not pre-specified, so it describes rather than tests [16].

Some values combine the leagues of several sports. They come from random-effects models, which give a combined value and the spread of the leagues around it [17]. They were fitted after the split's tests, so their values describe rather than test.

2.5 Sample size estimation and justification

The sample could not be chosen, so no size was estimated in advance. Every league that the sources cover deeply enough to carry the measures was read. In football, fourteen of the source's thirty-one top leagues have seasons under three substitutions, so its coverage sets the sample (Supplementary Section S1). That is a justification by the whole available population [18, 19]. For such a sample, this journal asks for the smallest effect each test could detect [20].

For an order, the smallest effect of interest is a rank correlation of 0.5. Below it, the order in one season accounts for less than a quarter of the differences in a later one.

To detect a link, a test needs a 0.80 chance of passing at a one-sided P below 0.05. Twelve leagues detect only a true rank correlation of 0.69 or more. Fourteen leagues detect 0.65 or more. So the order and explanation tests could detect only links well above the smallest of interest. The largest test of coaches who moved detects 0.20 or more. These values come from exact counts up to nine leagues and from simulation beyond, in Python, numpy and SciPy (Supplementary Section S1).

3 Results

We first checked the usual verdict, which judges each rule by its average effect. By that measure, each rule did what it was written to do (Table 2). The NHL, North America's top ice hockey league, brought in three-on-three overtime to settle more tied games before a shootout [21]. In the European leagues that followed, it multiplied the odds of settling a tied game before a shootout by 1.71. FIBA, basketball's world body, moved its three-point line out in 2010, to bring the world's rules together [22]. The move cut each league's share of shots from three by about 9.9 per cent of that share. Baseball's pitch clocks were written to speed up the game [23]. They shortened games in all four leagues tested (Supplementary Section S5).

3.1 Football's leagues kept their order through the change

Under three substitutions football's leagues sat close to their limit, and already in an order (Figure 1). A team made about 2.85 substitutions a match. The fourteen leagues' order over 2015–16 to 2018–19 was much the same as in their first season under five (rank correlation 0.89). This is a description, read after the change. A prediction written before we collected any records found the same in twelve European second tiers. Their order under three predicted their order under five (rank correlation 0.73, one-sided $P = 0.0041$).

Under five, each league added its own amount. Over 17 leagues, three of them outside Europe, the rise ran from 1.25 substitutions per team-match in Scotland to 1.70 in Brazil (Supplementary Section S2). But the leagues kept their order. In the main pre-specified test, the order of 12 European top leagues in their first season under five predicted their order in 2025–26 (rank correlation 0.73, one-sided $P = 0.0033$). Later, as the leagues climbed towards the new limit, their gaps narrowed while their order held (Supplementary Section S2).

We tested four material explanations of the order: money, rotation of the starting eleven, the value of the bench and absences from matchday squads. None showed a strong link to the order. With fourteen leagues, these tests could find only a strong link. In all, 51 tests read these and other explanations. After a correction for the number of tests, only one passed, and it largely restates how many substitutions a league makes (Supplementary Section S3).

3.2 Leagues in other sports kept their order in most tests

In most tests the leagues of other sports also kept their order in how widely they used their players. Baseball shows it best. A prediction for eight leagues was written before their 2026 seasons were read. Every league kept its rank in pitchers used per team and game. FIBA made a second change in 2014, when it shortened the shot clock's reset after an offensive rebound. Across both changes, basketball's leagues kept their order in the share of a team's minutes played by its five busiest players. The volleyball leagues kept their order in the number of players used, and cricket's thirteen T20 leagues in how many bowlers a side uses. Of the eight pre-specified tests of how widely leagues used their players, five held.

Some orders were lost. Basketball's pace, a team's possessions per forty minutes, lost its order when the 2014 change drew the leagues together. Two other tests, on when ice hockey teams pull their goalie and on baseball's pitchers in the split record, failed for a different reason. Each read only four leagues, where only a perfect order can pass (Supplementary Section S4).

3.3 A league's place persists in every sport measured, about as firmly as its scoring

In every sport measured, a league kept most of its place among its sport's leagues from one season to the next (Figure 2). The persistence of basketball, volleyball, ice hockey and baseball is consistent with one value, 0.969 a season. That value rests mostly on basketball's leagues. In basketball, about

78 per cent of a league's place is permanent, and lasted through nineteen seasons (Supplementary Section S6). Every prediction that this value would fit later seasons and new leagues held. The new leagues included the NBA and the WNBA, North America's top men's and women's leagues.

Other league quantities unrelated to rule use persisted almost as strongly. Goals, points and runs kept 0.93 to 0.96 of their place a season. So leagues keep their character in general [9, 24], and their habit is part of it.

3.4 Coaches carry part of a team's distance from its league

Coaches who moved carried part of their old team's distance from its league (Figure 3). The design follows coaches who move, as studies of managerial style do in firms and in football [25, 26]. Football managers carried their distance when they moved between clubs of one league, and when they changed league. Basketball's coaches carried it too, and a prediction on their moves since 2019 held. Baseball's pre-specified test, on its few moves, failed, so baseball is inconclusive. Combined over the three sports, the carry was a rank correlation of 0.30 (Supplementary Section S5).

The carry follows the coach, not the club's choice of coach. In football, a coach's distance at the old club correlated only 0.08 with the new club's own distance in the season before. Allowing for that earlier distance, the carry was 0.26 in football and 0.31 in basketball. A model that separates managers from their clubs found a part that is the coach's own: about 0.13 substitutions a match, or 4 a season (Supplementary Section S5).

4 Discussion

Each league used its rules in its own way, and kept doing so. In every sport measured, a league stayed near its place among its sport's leagues from season to season. Football's leagues carried the order they had under three substitutions through the change to five. We wrote eighteen predictions before their records were read, and thirteen held.

In sport science, a rule that gives room without saying how to use it is a task constraint [27]. Loosening a constraint widens the choices that work, and players and coaches settle among them [28, 29, 30]. In this view a competition organises itself [31]. Football's leagues show this at the scale of whole leagues. The limit of three held them close to it. When teams are held at a limit, the count shows only that they wanted at least that many [32]. But the order beneath the limit was already there. When the limit rose, each league settled at a level of its own below the new limit. So the new room mostly showed differences that the old limit had squeezed.

4.1 Practical applications

A rule-maker weighing a change can expect each league to use it in its own way, and to keep its place for several seasons. The spread across leagues and seasons was 0.39 substitutions per team-match. Over a season that is about 13 substitutions per club, and 289 minutes of play given to substitutes rather than starters. As the leagues neared the new limit, their gaps narrowed while their order

held. So the first seasons after a change overstate the gaps between the leagues, and cannot show how long each keeps its place.

Before a change, measure each league's use of the rule over at least three seasons, from records the league already keeps. After the change, compare each league with its own seasons before it, not with the average of all leagues. Judge the change over several seasons. Each league responds in its own way. So studies that combine leagues to estimate a rule's effect on injuries or running should allow for each league's response.

A club appointing a coach appoints part of how its team uses the rules. A coach's record at earlier clubs shows part of what to expect.

4.2 Relation to other work

The closest study uses the same rule change to study coaches' decisions [33]. It asks what the extra substitutions did within matches. This paper asks which leagues made them, and whether they kept doing so. A comparison of four of these leagues found that their results adapted alike to the change [6]. Here, their use of substitutions did not.

Several of these results first appeared in the first version of a preprint on nine sports [34]. This paper is that preprint's second version. The first version also claimed a law linking a league's load management to its timing. This version drops it: the two measures agree because of how they are built (Supplementary Section S7).

4.3 Limitations

No test here shows cause and effect. Every quantity describes how playing time was recorded, and a coach's reasons enter only where a team sheet records them. Most tests rest on four to 17 leagues, so they could detect only a strong link. The predictions on persistence would have failed only at a much weaker persistence, so their passes say little (Supplementary Section S6). No study register holds the pre-specified tests, so their dates rest on the project's repository (Supplementary Section S4). When football's main order test was written down, the seasons under five of France, Germany, Italy and Spain had already been read (Supplementary Section S2). The football records come from one public source, checked against an independent record only for some leagues and seasons (Supplementary Section S1). In 2020–21 and 2021–22 many records were incomplete. The rule change came with a pandemic. But England, which kept three substitutions through the crowded season of 2020–21, did not use more of them (Supplementary Section S7). Each league also sets rules of its own that were not measured in the other sports, such as roster sizes and limits on foreign players. These alone could keep a league in its place. Two predictions await the records of 2026–27 (Table 2; Supplementary Section S4). The coach's carry rests on managers who moved, and estimates from few movers can be distorted [35]. Supplementary Section S5 discusses the two published criticisms of the design [36, 37].

5 Conclusion

A speed limit sets how fast drivers may go, and each road settles at its own speed below it. Football's substitution limit worked the same way. When it rose, each league settled at its own level below the new limit, in the order the leagues already had. In every sport measured, a league's place persisted from season to season. We wrote predictions before their records were read, and most held. A rule sets the room, and each league uses that room in its own way. So a rule change should be judged league by league, against each league's own seasons before it, and over several seasons.

Data availability

All result tables, the analysis scripts and the test suite that rebuilds every number in this paper are available at `https://github.com/Gustolandia/team-sport-allocation-law` and in the archived record at DOI [to be reserved]. The match records come from third-party sources whose terms permit research use and not redistribution. The result tables are the minimum dataset needed to verify every claim. None of them names a player.

Disclosure of interest

The author reports there are no competing interests to declare.

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Declaration of generative AI use

Anthropic's Claude, a generative AI assistant, was used under the author's direction to write and check the analysis scripts, search the literature and edit the text. The version used for this revision was Claude Opus 5.5, through the Claude Code tool. It was used because a single author could not otherwise build and check analyses of this size. The study design, the claims and their interpretation are the author's. Every reported quantity is computed from the committed data by the archived scripts and was verified by the author, who takes full responsibility for the content.

References

[1] Arias JL, Argudo FM, Alonso JI. Review of rule modification in sport. Journal of Sports Science and Medicine. 2011;10(1):1-8.

[2] Rennie G, Chesson L, Weaving D, Jones B. The effects of rule changes in football-code team sports: a systematic review. Science and Medicine in Football. 2025;9(3):199-212. doi:10.1080/24733938.2024.2375752.

[3] Martinho DV, Gonzalo-Skok O, Gómez Ruano MÁ, Clemente FM, Gouveia ÉR, Nunes NA, et al. Substitutions in elite soccer: a scoping review. International Journal of Performance Analysis in Sport. 2026:1-25. Published online 10 March 2026. doi:10.1080/24748668.2026.2635275.

[4] Ayabe M, Sunami S, Kumahara H, Ishizaki S. Effects of substitute allowance on match activity characteristics in Japanese professional football across 2019, 2020, and 2021 seasons. Journal of Sports Sciences. 2022;40(23):2654-60. doi:10.1080/02640414.2023.2182878.

[5] Dilger A, Vischer L. Effects of the rule change from three to five substitutions in the Bundesliga. International Journal of Sport Finance. 2024;19(4):198-207. doi:10.32731/ijsf.194.112024.01.

[6] Zhang H, Mathes J, Garnica Caparrós M, Memmert D. Assessing the impact of substitution rule changes in European football leagues. International Journal of Sports Science & Coaching. 2025. Published online 21 February 2025. doi:10.1177/17479541251316162.

[7] The International Football Association Board. Laws of the Game 2022/23. Zurich: The International Football Association Board; 2022.

[8] Meyer J, Klatt S. Additional substitutions in elite European football. International Journal of Sports Science & Coaching. 2024;19(2):769-78. doi:10.1177/17479541231164090.

[9] Pollard R. Worldwide regional variations in home advantage in association football. Journal of Sports Sciences. 2006;24(3):231-40. doi:10.1080/02640410500141836.

[10] Sapp RM, Spangenburg EE, Hagberg JM. Trends in aggressive play and refereeing among the top five European soccer leagues. Journal of Sports Sciences. 2018;36(12):1346-54. doi:10.1080/02640414.2017.1377911.

[11] Caley M. Substitutes, Part II: clubs, leagues and managers; 2024. Expecting Goals, 8 February 2024. Available from: https://www.expectinggoals.com/p/substitutes-part-ii-clubs-leagues.

[12] Segar D. How substitute trends compare across Europe's top five leagues; 2024. Opta Analyst, 28 March 2024. Available from: https://theanalyst.com/articles/how-substitute-trends-compare-across-europe-top-five-leagues.

[13] Cariboo D. transfermarkt-datasets: clean, structured football data built from Transfermarkt; 2026. GitHub repository, commit af8c44a, snapshot of 3 August 2026, CC0-1.0. Available from: https://github.com/dcaribou/transfermarkt-datasets.

[14] Cricsheet. Ball-by-ball data for limited-overs cricket; 2026. cricsheet.org, archive of all matches in JSON, downloaded 6 September 2026.

[15] Trinity College Dublin. Guidance for using the Research Ethics Application Management System (REAMS): making an application; 2022. Institutional guidance, pages 5 and 27–28. Accessed 25 September 2026. Available from: https://www.tcd.ie/research/support/assets/pdf/Top%20Guidance_Simplified-31st%20Oct.pdf.

[16] Ditroilo M, Mesquida C, Abt G, Lakens D. Exploratory research in sport and exercise science: Perceptions, challenges, and recommendations. Journal of Sports Sciences. 2025;43(12):1108-20. doi:10.1080/02640414.2025.2486871.

[17] Riley RD, Higgins JPT, Deeks JJ. Interpretation of random effects meta-analyses. BMJ. 2011;342:d549. doi:10.1136/bmj.d549.

[18] Lakens D. Sample size justification. Collabra: Psychology. 2022;8(1):33267. doi:10.1525/collabra.33267.

[19] Wagnsson S, Lundkvist E. Scoreless but still scoring: A population-based observational study of removing league tables and playoffs in children's soccer. Journal of Sports Sciences. 2026;44(19):2709-19. doi:10.1080/02640414.2026.2713886.

[20] Abt G, Boreham C, Davison G, Jackson R, Jobson S, Wallace E, et al. Sample size estimation revisited. Journal of Sports Sciences. 2025;43(21):2511-6. doi:10.1080/02640414.2025.2499403.

[21] Rosen D. Board of Governors OKs 3-on-3 OT, coach's challenge; 2015. NHL.com, 24 June 2015; Internet Archive copy of 31 October 2015 (accessed 15 October 2026). Available from: https://web.archive.org/web/20151031130948/http://www.nhl.com/ice/news.htm?id=771982.

[22] International Basketball Federation (FIBA). FIBA approves historic rule changes; 2008. Press release, Geneva, 26 April 2008, as republished by Basketball Manitoba on 29 April 2008 (accessed 15 October 2026). Available from: https://www.basketballmanitoba.ca/2008/04/fiba-approves-historic-rule-changes.html.

[23] Major League Baseball. MLB announces rule changes for 2023 season; 2022. Press release, 9 September 2022 (accessed 15 October 2026). Available from: https://www.mlb.com/press-release/press-release-mlb-announces-rule-changes-for-2023-season.

[24] Fort R, Lee YH. Structural change, competitive balance, and the rest of the major leagues. Economic Inquiry. 2007;45(3):519-32. doi:10.1111/j.1465-7295.2007.00026.x.

[25] Bertrand M, Schoar A. Managing with Style: The Effect of Managers on Firm Policies. The Quarterly Journal of Economics. 2003;118(4):1169-208. doi:10.1162/003355303322552775.

[26] Muehlheusser G, Schneemann S, Sliwka D, Wallmeier N. The contribution of managers to organizational success: evidence from German soccer. Journal of Sports Economics. 2018;19(6):786-819. doi:10.1177/1527002516674760.

[27] Newell KM. Constraints on the development of coordination. In: Wade MG, Whiting HTA, editors. Motor Development in Children: Aspects of Coordination and Control. Dordrecht: Martinus Nijhoff; 1986. p. 341-60. doi:10.1007/978-94-009-4460-2_19.

[28] Renshaw I, Davids K, Newcombe D, Roberts W. The Constraints-Led Approach: Principles for Sports Coaching and Practice Design. Abingdon: Routledge; 2019. doi:10.4324/9781315102351.

[29] Balagué N, Pol R, Torrents C, Ric A, Hristovski R. On the relatedness and nestedness of constraints. Sports Medicine - Open. 2019;5(1):6. doi:10.1186/s40798-019-0178-z.

[30] Ashford M, Burke K, Barrell D, Abraham A, Poolton J. The impact of rule modifications on player behaviour in a talent identification and development environment: A case study of the Rugby Football Union's Wellington Academy Rugby Festival. Journal of Sports Sciences. 2020;38(23):2670-6. doi:10.1080/02640414.2020.1795559.

[31] McGarry T, Anderson DI, Wallace SA, Hughes MD, Franks IM. Sport competition as a dynamical self-organizing system. Journal of Sports Sciences. 2002;20(10):771-81. doi:10.1080/026404102320675620.

[32] Kleven HJ. Bunching. Annual Review of Economics. 2016;8(1):435-64. doi:10.1146/annurev-economics-080315-015234.

[33] Cermak L. The Negative Productivity of Strategic Flexibility: Evidence from a Natural Experiment in Football; 2026. SSRN working paper. doi:10.2139/ssrn.6983270.

[34] Ricou GP. One equation allocates playing time in competitive team sports: the rulebook sets the arithmetic and the league sets the habit; 2026. arXiv:2609.05584v1 [physics.soc-ph].

[35] Andrews MJ, Gill L, Schank T, Upward R. High wage workers and low wage firms: negative assortative matching or limited mobility bias? Journal of the Royal Statistical Society: Series A (Statistics in Society). 2008;171(3):673-97. doi:10.1111/j.1467-985X.2007.00533.x.

[36] Fee CE, Hadlock CJ, Pierce JR. Managers with and without Style: Evidence Using Exogenous Variation. The Review of Financial Studies. 2013;26(3):567-601. doi:10.1093/rfs/hhs131.

[37] Jarosiewicz VE, Ross DG. Revisiting Managerial "Style": The Replicability and Falsifiability of Manager Fixed Effects for Firm Policies. Strategic Management Journal. 2023;44(3):858-86. doi:10.1002/smj.3439.

Table 1: The seven sports, the rule change studied in each, and every record the analyses read. The counts leave out later seasons that an analysis read only as league means. They also leave out the further football leagues that only the explanation models read (Supplementary Section S1). Each league-season, one league's season, is counted once, at the largest count any analysis read. A change splits a sport's seasons into those under the old rule and those under the new. Supplementary Tables S1 and S2 give each league's seasons. For football's men's leagues, the number in brackets gives the fourteen European leagues whose substitutions are counted. Football's other leagues enter the timing measures (Supplementary Section S7). Seasons named by one year are played within a calendar year. Cricket's row covers a rule that did not change.

Sport	Rule change read	Seasons	Leagues	Records
Football, men	substitution limit raised from three to five, between 2019–20 and 2022–23 by league	2015–16 to 2025–26	21 (14)	47,937 matches
Football, women	the same limit, read for the timing of substitutions	2023–24 to 2025	7	1,130 matches
Basketball	three-point line moved out, 2010–11; shot clock reset after an offensive rebound shortened, 2014–15	2006–07 to 2025	22	55,051 games
Volleyball	more players allowed on the match sheet, at dates that differ by league	2006–07 to 2025–26	8	21,297 games
Handball	a third team time-out, from January 2012	2009–10 to 2019–20	5	6,784 games
Ice hockey	tied games settled by three-on-three overtime before a shootout, at dates that differ by league	2011–12 to 2025–26	12	44,312 games
Baseball	a pitch clock, enforced from 2023 to 2025 by league	2008 to 2025	12	76,872 games
Cricket (T20)	none: at most a fifth of an innings per bowler throughout (an Impact Player rule in one league from 2023)	2019 to 2025	16	3,583 innings
Six other sports			75	207,899 games and innings
Seven sports			103	256,966 matches, games and innings

Table 2: Every pre-specified test of the paper's claims and checks, failures included, in three blocks. Of eighteen predictions on new records, thirteen held. The predictions on new records were written down before any of their records was read. The split record's tests were saved before half B was read. Supplementary Section S4 dates football's own tests, and Supplementary Section S7 describes its timing tests. The letter after each block's name begins the labels of its tests in the supplement. The smallest attainable P is the smallest P a test can reach with its number of leagues, or with the number of simulations used to compute it. A dash marks a test with no such floor. Power is a rank test's chance of passing when the true rank correlation is 0.7, with a one-sided P below 0.05 to pass. [a]The row includes a test that could hardly pass, or hardly fail, with its leagues.

Claim	Asked in	Leagues	Smallest attainable P	Power at 0.7	Held / asked
predictions on new records (G)					
One constant per rule (the 2010 line; three-on-three overtime)	basketball; ice hockey	1–2	–	–	1 of 2
One proportion of the share before, with a spread (the 2010 line)	basketball	6	–	–	1 of 1
One constant per second of clock (the first pitch clocks)	baseball	6	–	–	0 of 1
A league keeps its place (basketball)	basketball	7	2.0×10^{-4}	0.49	2 of 4
The order holds over more leagues (ice hockey, baseball, cricket)[a]	ice hockey; baseball; cricket	6–13	2.5×10^{-5} to 0.0014	0.32 to 0.85	3 of 4
One persistence a season across sports[a]	ice hockey; basketball; cricket	5–13	–	–	4 of 4
One carry across sports (the coach)	basketball	3	1.0×10^{-4}	1.00	1 of 1
Football's rules in new seasons and new leagues (written down in this revision)[a]	football	14–30	1.0×10^{-5}	–	1 of 1 (2 not yet read)
split record (R)					
A rule moves what it was written to move	basketball; ice hockey; baseball	4–6	–	–	3 of 3
The order of the leagues persists[a]	basketball; volleyball; ice hockey; baseball	4–8	2.5×10^{-5} to 0.042	0.22 to 0.60	4 of 8
A tightening draws the leagues together	basketball; baseball	4–7	5.0×10^{-4}	–	1 of 3
A loosened limit is used more, and lets the leagues differ	handball; volleyball; basketball; ice hockey	1–8	5.0×10^{-4}	–	4 of 6
A climb slows and levels off	handball; ice hockey; baseball	1–4	–	–	0 of 3
The habit before a change predicts the response	volleyball	5	–	–	0 of 1
The coach carries part of a team's distance from its league	basketball	3	1.0×10^{-4}	1.00	2 of 2
football's own pre-specified tests (C)					
The order persists across the five-substitution era (the order test)	football	12	2.1×10^{-9}	0.81	1 of 1
A material explanation accounts for the order (money, rotation, bench value, absences)	football	12–14	4.8×10^{-14} to 1.6×10^{-10}	0.85 to 0.92	0 of 4
A further explanation accounts for the order (culture; the league's own increment; two batteries)	football	13–31	1.2×10^{-34} to 1.6×10^{-10}	0.85 to 0.94	0 of 8
The coach's carry (the movers tests)	football; baseball	1–14	–	0.97 to 1.00	5 of 7
Timing before the change (the pre-change tests)[a]	football	8–12	2.5×10^{-5}	0.60	1 of 4
The timing slope on separate halves of the matches[a]	football	12	–	–	1 of 1
Resting between matches against timing within them[a]	football	8	2.5×10^{-5}	0.60	0 of 1

Figure captions

Figure 1. The fourteen football leagues carried the order they had under three substitutions through the change to five. (a) Substitutions per team-match in the fourteen European leagues, season by season, under the limit each season's regulations set. Each league is named by its country. The three highest leagues and the three lowest in 2025–26 are labelled, in the darker greys. The other eight are in light grey. England, dashed, used five in the restart of 2019–20 and went back to three for 2020–21 and 2021–22. The band marks 2019–20, which eight leagues split at the date of their change. Belgium's 2020–21 and Scotland's 2021–22 are split the same way, with a small grey box behind each part. Each part of a split season is drawn as its own point. Each mean is uncertain by about half a marker's height under three, and by about a marker's height under five (one standard error). Open markers are the fifteen league-seasons whose records were incomplete for more than a tenth of their team-matches (Supplementary Section S1). Several overlap in 2021–22. Ukraine's 2021–22 is not drawn, because almost none of its team-matches were kept. (b) The same counts as a share of each season's limit. (c) Each league's mean under three, over 2015–16 to 2018–19, against its mean under five, over every season under five. Each league keeps its marker from (a), and the leagues named in (a) are named here. The rank correlation ρ between the two means is printed with its 95% interval. It is description, read after the change. Open points are Denmark and England, which are outside the pre-specified test (Supplementary Section S2).

Figure 2. In every sport measured, a league's place persists, about as firmly as its scoring. (a) The persistence ϕ, the share of its place a league keeps from one season to the next, with its 95% interval. Beside it are the leagues and pairs of seasons behind it, and each sport's weight in the pooled value, which combines the sports. Squares are sports, with their measures together, and circles are single measures. The open circle counts Japan's volleyball sets as played (Supplementary Section S4). The diamond is the pooled value over basketball, volleyball, ice hockey and baseball. Open squares are predictions, written down before their records were read, and they are not in the pool. Grey points are scoring, for comparison: goals, points and runs. The dashed line is the pooled value, and the dotted line a persistence of one. An interval that reaches past one comes from few seasons. The scale at the top gives the half-life under one rate: the number of seasons until a league would lose half of its place. (b) Basketball, which carries most of the pooled weight. Each point is the mean correlation of the leagues' places k seasons apart, with the chance swings of single games left in. Neighbouring values of k are grouped, and each point has its 95% interval over leagues. Each line is the mean correlation that a model predicts for the same pairs, with those swings left in. The solid line has one rate, basketball's ϕ. The dashed line has a permanent part, about 78 per cent of a league's place, plus a part that decays. It is fitted to the leagues' seasons, not to these means. It runs a little above the points, but it follows their levelling off, which one rate cannot.

Figure 3. Coaches carry part of a team's distance from its league, and every reading in three sports points the same way. Each row gives the rank correlation ρ between two distances from the league, with its 95% interval. One is the distance of the team a coach or manager left, and the other

is the new team's. Beside it are the slope of a straight-line fit, with its interval, and the number of moves. Each sport has its own measure: substitutions a match in football, the five busiest players' share of minutes in basketball, and the players used beyond the starters in baseball. Black points, filled or open, enter the pooled value, which combines the sports, with their weight at right. The open point, the managers of MLB, the top baseball league of the United States and Canada, is inconclusive. Grey points are other readings of the same carry (Supplementary Section S5). The rows marked "given the destination" allow for the new club's own distance in the season before. Basketball's readings of every game and of half the games share most of their moves. The black diamond is the pooled value over the three sports, with its 95% interval, and the dashed line marks it. Neither ρ nor the slope is corrected for the chance swings of single games. Those swings blur any link, so both understate the carry.

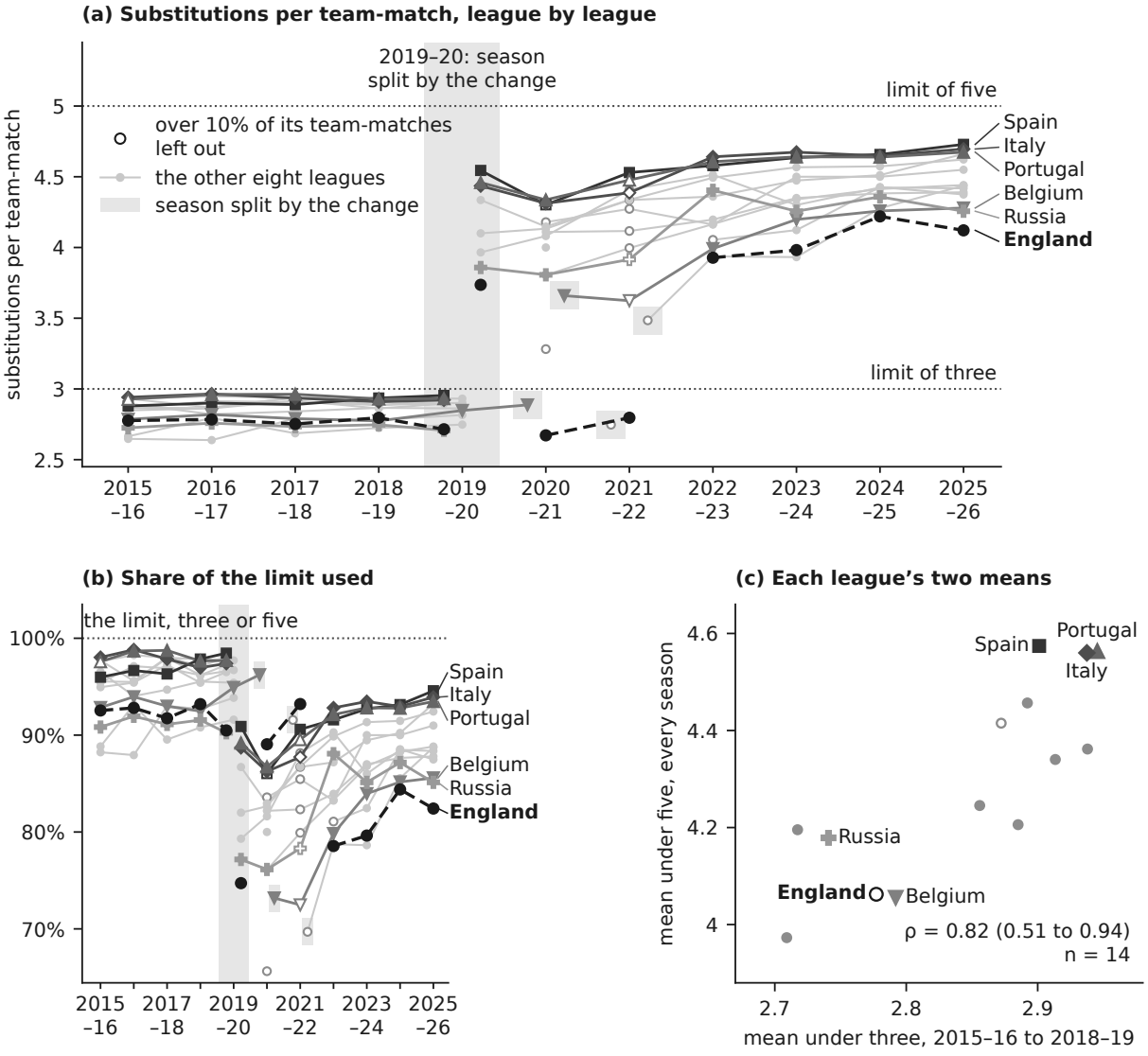

(a) Substitutions per team-match, league by league
2019–20: season split by the change
limit of five
limit of three
over 10% of its team-matches left out
the other eight leagues
season split by the change
substitutions per team-match
5
4.5
4
3.5
3
2.5
Spain
Italy
Portugal
Belgium
Russia
England
2015 –16
2016 –17
2017 –18
2018 –19
2019 –20
2020 –21
2021 –22
2022 –23
2023 –24
2024 –25
2025 –26
(b) Share of the limit used
the limit, three or five
100%
90%
80%
70%
Spain
Italy
Portugal
Belgium
Russia
England
2015 –16
2017 –18
2019 –20
2021 –22
2023 –24
2025 –26
(c) Each league's two means
mean under five, every season
4.6
4.4
4.2
4
Spain
Portugal
Italy
Russia
England
Belgium
ρ = 0.82 (0.51 to 0.94)
n = 14
2.7
2.8
2.9
mean under three, 2015–16 to 2018–19

Figure 1

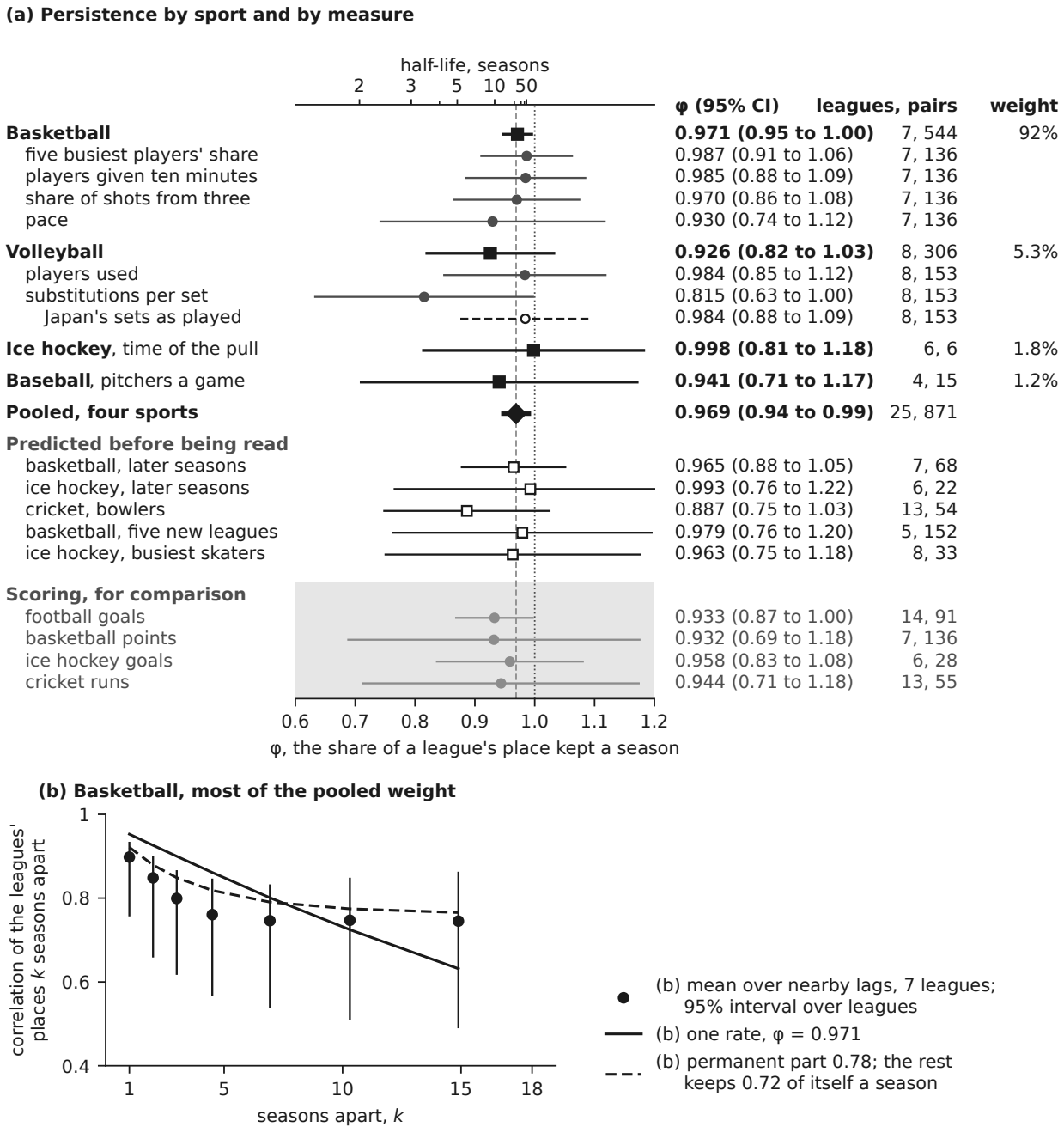

(a) Persistence by sport and by measure
half-life, seasons
2
3
5
10
50
φ (95% CI)
leagues, pairs
weight
Basketball
0.971 (0.95 to 1.00)
7, 544
92%
five busiest players' share
0.987 (0.91 to 1.06)
7, 136
players given ten minutes
0.985 (0.88 to 1.09)
7, 136
share of shots from three
0.970 (0.86 to 1.08)
7, 136
pace
0.930 (0.74 to 1.12)
7, 136
Volleyball
0.926 (0.82 to 1.03)
8, 306
5.3%
players used
0.984 (0.85 to 1.12)
8, 153
substitutions per set
0.815 (0.63 to 1.00)
8, 153
Japan's sets as played
0.984 (0.88 to 1.09)
8, 153
Ice hockey, time of the pull
0.998 (0.81 to 1.18)
6, 6
1.8%
Baseball, pitchers a game
0.941 (0.71 to 1.17)
4, 15
1.2%
Pooled, four sports
0.969 (0.94 to 0.99)
25, 871
Predicted before being read
basketball, later seasons
0.965 (0.88 to 1.05)
7, 68
ice hockey, later seasons
0.993 (0.76 to 1.22)
6, 22
cricket, bowlers
0.887 (0.75 to 1.03)
13, 54
basketball, five new leagues
0.979 (0.76 to 1.20)
5, 152
ice hockey, busiest skaters
0.963 (0.75 to 1.18)
8, 33
Scoring, for comparison
football goals
0.933 (0.87 to 1.00)
14, 91
basketball points
0.932 (0.69 to 1.18)
7, 136
ice hockey goals
0.958 (0.83 to 1.08)
6, 28
cricket runs
0.944 (0.71 to 1.18)
13, 55
0.6
0.7
0.8
0.9
1.0
1.1
1.2
φ, the share of a league's place kept a season
(b) Basketball, most of the pooled weight
correlation of the leagues' places k seasons apart
1
0.8
0.6
0.4
1
5
10
15
18
seasons apart, k
(b) mean over nearby lags, 7 leagues; 95% interval over leagues
(b) one rate, φ = 0.971
(b) permanent part 0.78; the rest keeps 0.72 of itself a season

Figure 2

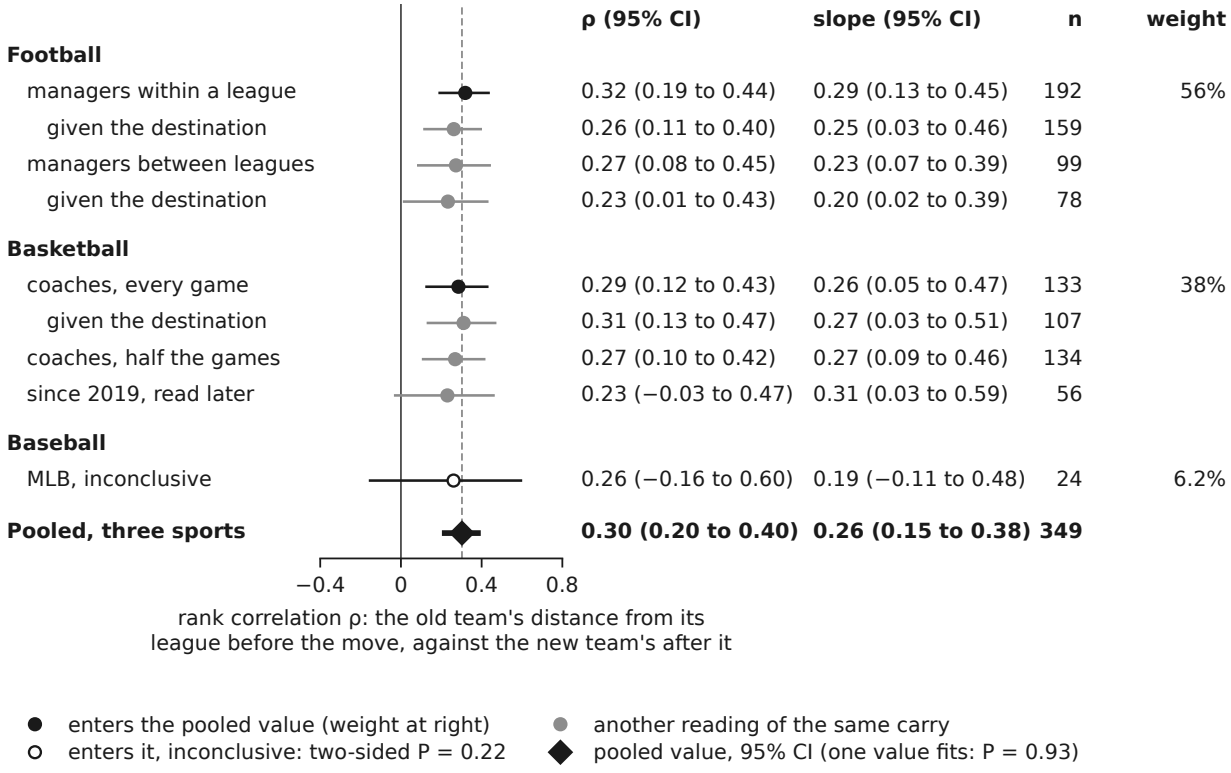

ρ (95% CI)
slope (95% CI)
n
weight
Football
managers within a league
0.32 (0.19 to 0.44)
0.29 (0.13 to 0.45)
192
56%
given the destination
0.26 (0.11 to 0.40)
0.25 (0.03 to 0.46)
159
managers between leagues
0.27 (0.08 to 0.45)
0.23 (0.07 to 0.39)
99
given the destination
0.23 (0.01 to 0.43)
0.20 (0.02 to 0.39)
78
Basketball
coaches, every game
0.29 (0.12 to 0.43)
0.26 (0.05 to 0.47)
133
38%
given the destination
0.31 (0.13 to 0.47)
0.27 (0.03 to 0.51)
107
coaches, half the games
0.27 (0.10 to 0.42)
0.27 (0.09 to 0.46)
134
since 2019, read later
0.23 (−0.03 to 0.47)
0.31 (0.03 to 0.59)
56
Baseball
MLB, inconclusive
0.26 (−0.16 to 0.60)
0.19 (−0.11 to 0.48)
24
6.2%
Pooled, three sports
0.30 (0.20 to 0.40)
0.26 (0.15 to 0.38)
349
−0.4
0
0.4
0.8
rank correlation ρ: the old team's distance from its league before the move, against the new team's after it
enters the pooled value (weight at right)
enters it, inconclusive: two-sided P = 0.22
another reading of the same carry
pooled value, 95% CI (one value fits: P = 0.93)

Figure 3